

Assignment – 06:

Problem Statement:

Write a C function to compute the factorial of a number with and without recursion.

Code:

```
#include <stdio.h>

int fact_iterative (int n)
{
    int fact=1;
    for (int i=1;i<=n;i++)
    {
        fact=fact*i;
    }
    return fact;
}

int fact_recursive (int n)
{
    if (n==0||n==1)
    {
        return 1;
    }
    else
    {
        return n*fact_recursive(n-1);
    }
}

int main()
{
    int num;
```

```
printf("Enter a number: ");
scanf("%d",&num);
if (num<0)
{
    printf("Factorial not defined for negative numbers\n");
}
else
{
    printf("Factorial (Without Recursion) = %d\n",fact_iterative(num));
    printf("Factorial (With Recursion) = %d\n",fact_recursive(num));
}
}
```

Input & Output:

```
Enter a number: -2
Factorial not defined for negative numbers
```

```
Enter a number: 0
Factorial (Without Recursion) = 1
Factorial (With Recursion) = 1
```

```
Enter a number: 4
Factorial (Without Recursion) = 24
Factorial (With Recursion) = 24
```

